# Sagar Saha

National Postdoctoral Fellow (NPDF), IISER Mohali
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## Research interests

Branch groups; groups acting on rooted trees; self-similar and automata groups; combinatorial and geometric group theory. Particular interests include non-contracting branch groups and rigid kernels.

## Academic appointments

**September 2026-present:** National Postdoctoral Fellow (NPDF), IISER Mohali. Working with Prof. Mahender Singh.

**April-September 2026:** Research Associate, IISER Thiruvananthapuram. Working with Prof. Viji Z. Thomas.

## Education

**2019-March 2026:** PhD in Mathematics, IIT Guwahati. Advisor: Prof. K. V. Krishna.

**2017-2019:** MSc in Mathematics and Scientific Computing, NIT Warangal.

**2014-2017:** BSc (Honours) in Mathematics, Dinhata College, University of North Bengal.

## Publications

S. Saha and K. V. Krishna. “A class of non-contracting branch groups with non-torsion rigid kernels.” *Journal of Algebra* **692** (2026), 379-400. arXiv:2501.04112.

S. Saha and K. V. Krishna. “A branch group in a class of non-contracting weakly regular branch groups.” *International Journal of Algebra and Computation* **34**(6) (2024), 901-918. arXiv:2311.06393.

## Fellowships and honours

National Postdoctoral Fellowship (NPDF), from September 2026.

NBHM Postdoctoral Fellowship, awarded April 2026.

Best Paper Presentation Award, ISTAA 2024, University of North Bengal.

Merit-cum-means Scholarship, NIT Warangal, 2017-2019.

Qualified GATE in Mathematics (2019), CSIR-UGC NET in Mathematical Sciences (2018), and IIT JAM in Mathematics (2017).

## Talks

ISTAA 2024, University of North Bengal: “A Branch Group in a Class of Non-Contracting Weakly Regular Branch Groups.”

ICSOA 2024, University of Kerala: “Certain Properties of a Family of Non-Contracting Automata Groups.”

Geometry in Groups (GiG), ICTS-TIFR, Bengaluru, 2024: lightning talk, “A Class of Non-Contracting Branch Groups of Exponential Growth.”

## Conference and workshop participation

GAGTA 2025, Stevens Institute of Technology, USA (online).

IWM Regional Workshop on Research and Opportunities, NIT Meghalaya, 2019.

## Teaching experience

Teaching Assistant, IIT Guwahati: MA101 (Single Variable Calculus and Multi-Variable Calculus); MA102 (Linear Algebra and Ordinary Differential Equations); MA521 (Modern Algebra); MA514 (Theory of Computation).

## Mathematical software and programming

GAP, Mathematica, MATLAB; C and C++.